

Pi 5 NOT BOOTING

Since this morning, one of my Raspberry Pi 5 is not booting.
I have 2 Raspberry Pi 5: the problematic one is a Pi 5 with 8Gb that I will identify as pi-broke I have another, functional Pi 5 with 2Gb that I will identify as pi-work, I used the latter as comparison tests mentioned below

1. Configuration

pi-broke was loading Ubuntu Server from a SD card.

2. Symptom

I removed all peripherals, left the SD card, and a USB-C for power. Plug in USB-C, the power LED stays solid red. Pressing the power button does nothing. The board never reaches green, never boots.

3. Tests

3.1 Swap Test

Swap the SD card to the working pi. It boots.
Note: the SD card holds the OS, not the bootloader. The bootloader lives in the board's onboard SPI EEPROM. A card swap tells you nothing about EEPROM state.

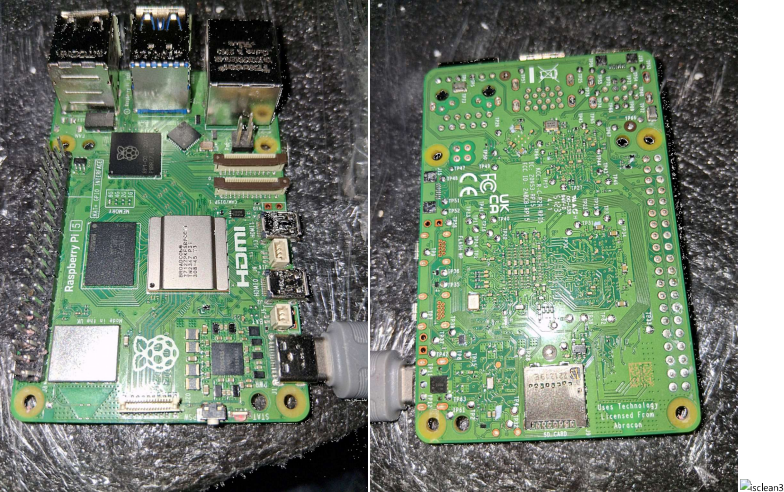
3.2 Hard Drain and Minimal Config

Clears a PMIC that latched into a protective state on an overcurrent or thermal event.

- 1. Unplug everything
- 2. Hold the power button ~10s to bleed residual charge.
- 3. Replug with no SD card and no peripherals.
- 4. Watch the LED.

3.3 Visual Inspection

Inspect both sides of the board for defects, damages or burn marks: It's all good, see pictures.



3.4 Test 5V Input Rail Resistance -- GOOD

Dead board. **USB-C unplugged, SD card out, nothing connected.** Resistance measurements are always done unpowered.
DMM in ohms mode, probe:

GND pin 2 (5V) to GND pin 6 (GND)

Compare against a known-good board.

Reading	Meaning
~24 kΩ (reference, known-good)	3.3V rail clean.

We get about 20 K Ohms on both.

3.5 Test D: 3.3V Rail Resistance - ERROR

Probing:

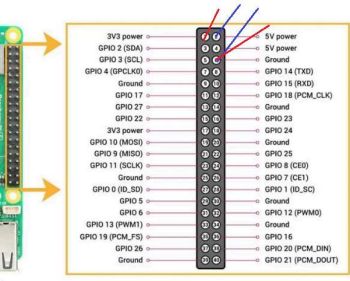
GND pin 1 (5V3) to GND pin 6 (GND)

Compare against a known-good board (pi-work)

Reading	Meaning
~24 kΩ (reference, known-good)	3.3V rail clean.
Large discrepancy vs good board, e.g. ~400 Ω	3.3V rail shorted or partially shorted. This is the fault.

There's a partial short (e.g. ~393 Ω vs ~24 kΩ good):

- Not a dead 0 Ω short, so the PMIC output FET is not fused closed.
- A resistive pull-down on the rail. Most likely a failed decoupling ceramic (MLCC) gone leaky or partially shorted, a classic post-thermal-cycle failure. Less likely, a 3.3V peripheral IC failed internally.
- Prognosis shifts from "scrap" to "potentially repairable," because a shorted cap is findable and removable.



5V RAIL RESISTANCE TEST
GPIO 2 to GPIO 6 ==> 14 Kilo Ohms
ON BOTH THE WORKING AND BROKE PI
3.3 RAIL RESISTANCE TEST
GPIO 1 to GPIO 6 ==> 400 Ohms on BROKE PI
19 Kilo Ohms on Working PI
3.3 RAIL LOOKS SHORTED